

NMC 2013 A-LEVEL PAPER 1 QUESTIONS

SECTION A : 5 marks each

1. Function $f(x)$ obeys the rule $f(x+1) = f(x) - f(x-1)$. Given $f(2) = 5$ and $f(1) = 2$, find $f(2009)$.

2. Find the value of $\sqrt{7-\sqrt{48}} + \sqrt{5-\sqrt{24}} + \sqrt{3-\sqrt{8}}$.

3. Evaluate $1! \cdot 3 - 2! \cdot 4 + 3! \cdot 5 - 4! \cdot 6 + \ldots - 2012! \cdot 2014 + 2013!$.

4. The solutions to the equation $x^2 + px + q = 0$ are cubes of the solutions of the equation $x^2 + mx + n = 0$. Write down an equation relating m, n and p.

5. An ice-cream van sells nine different flavours of ice-cream. A large group of children gathered by the van and each of them bought a double-scoop con with two different flavours of ice-cream. If none of the children chose the same two flavours, What is the largest possible number of children in the group?

6. Jonathan never lies on Monday, Tuesday, Wednesday and Thursday. David always tells the truth on Monday, Friday, Saturday and Sunday. On the rest of the days, they may tell the truth or they may lie. Both say they lied yesterday. What day is today?

7. Each square in the partially filled grid shown contains a number from 1 to 25, each number appearing exactly once. The numbers in each row and each column add up to 65. What is the value of $x + y$?

	7	25	18	
	21	19	y	
22	20			9
16			10	23
	x	6	24	

8. Given the system of equations.

$$\begin{aligned} x+y &= 2m^2 \\ y+z &= 6m \\ x+z &= 2 \end{aligned}$$

where m is a real number. For what values of m is $x \leq y \leq z$?

9. Jane and Sarah run at different but constant speeds. They ran two races on a track that measured $100m$ from start to finish. In the first race, when Jane crossed the finish line, Sarah was $5m$ behind. In the second race, Jane started $5m$ behind the original start line and they ran at the same speeds as in the first race. What was the outcome of the second race?

10. Find all functions $f : \mathbb{R}^+ \longrightarrow \mathbb{R}$ such that $xy\left(3f(\frac{x}{y}) + 5f(\frac{y}{x})\right) = 3x^2 + 5y^2$. where $\mathbb{R}^+$ consists of positive real numbers and $\mathbb{R}$ consists of all real numbers.

SECTION B : 10 **marks each**

11. Twelve different balls are to be packed into 4 different boxes so that each box contains 3 balls. Find the number of ways of packing the balls in this manner.

12. Each of 52 cards has an integer written on it (repetitions allowed). The numbers on any three consecutive cards add up to 20. If the number on the first card is 2 and the number on the ninth card is 8. What is the number on the 52nd card?

13. Let a, b and c be distinct positive real numbers that form a geometric sequence. Let $b - a$ be a square of a natural number n and

$$\log_6 a + \log_6 b + \log_6 c = 6$$

Find the value of $a + b + c$.

14. Let P, Q and R be real numbers such that

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots + \frac{n}{2^n} = \frac{Pn+Q}{2^n} + R$$

holds for every positive integer n. What is the value of $P + Q + R$?

15. In the figure shown, $< ABC$ and $< BDA$ are both right angles. If $v + w = 35$ and $x + y = 37$. What is the value of $y + 8$?

NMC 2013 A-LEVEL PAPER 1 SOLUTIONS

SECTION A

1. Given $f(2) = 5$ and $f(1) = 2$, $f(3) = f(2) - f(1) = 3$, $f(4) = f(3) - f(2) = -2$, $f(5) = -5$, $f(6) = -3$, $f(7) = 2(= f(1))$, $f(8) = 5(= f(2))$. The function $f(x)$ is periodic , with period 6. $2013 \bmod 6 = 3$ (Since $2013 = 6 \times 335 + 3$.) Therefore $f(2013) = f(3) = 3$.

2. Note that $7 - \sqrt{48} = (2 - \sqrt{3})^2$, $\quad 5 - \sqrt{24} = (\sqrt{3} - \sqrt{2})^2$, $\quad 3 - \sqrt{8} = (\sqrt{2} - 1)^2$.

   $$\sqrt{7 - \sqrt{48}} + \sqrt{5 - \sqrt{24}} + \sqrt{3 - \sqrt{8}}. = \sqrt{(2 - \sqrt{3})^2} + \sqrt{(\sqrt{3} - \sqrt{2})^2} + \sqrt{(\sqrt{2} - 1)^2}$$

   $$= 2 - \sqrt{3} + \sqrt{3} - \sqrt{2} + \sqrt{2} - 1 = 1$$

3. Note that $n! = (n+1)! = n!(n+2)$.
   Using this observation, we get
   $1! \cdot 3 - 2! \cdot 4 + 3! \cdot 5 - 4! \cdot 6 + \ldots - 2012! \cdot 2014 + 2013! = 1! + 2! - (2! + 3!) - \cdots - 2012! - 2013! + 2013! = 1$

4. Let x_1, x_2 be solutions of $x^2 + mx + n = 0$, then x_1^3 and x_2^3 are solutions to $x^2 + px + q = 0$. This implies $x_1 + x_2 = -m$ and $x_1 x_2 = n \Longrightarrow x_1^3 + x_2^3 = -p$ and $x_1^3 x_2^3 = q$.
   From $(x_1 + x_2)^3 = x_1^3 + x_2^3 + 3x_1^2 x_2 + 3x_1 x_2^2 = (x_1 + x_2)^3 + 3x_1 x_2 (x_1 + x_2)$
   $(-m)^3 = -p + 3n(-m)$. Therefore $m^3 = p + 3nm$.

5. For the first scoop, a child has a free choice that is 9 options. For the second scoop, there are only 8 choices left because the choice of say, stawberry and vanilla is identical to that of vanilla and strawberry. There are $\frac{9 \times 8}{2} = \frac{72}{2} = 36$ choices of flavours. Hence the largest possible number of children in the group is 36.

6. Jonathan never lies on Monday,Tuesday, Wednesday and Thursday. we are not sure whether he lies on Friday, Saturday or sunday. David never lies on Monday, friday, Saturday, Sunday. we are not sure whether he lies on Tuesday, Wednesday or Thursday.
   Suppose today is saturday, then yesterday was Friday on which day we are not sure whether Jonathan lies or not and david never lies on Friday which is a contradiction.
   By continuous elimination gives today as Friday.

7.

	7	25	18	
	21	19	y	
22	20	16-x	x-2	9
16	17-x	x-1	10	23
	x	6	24	

   Using the second column, we have the unfilled box as $17 - x$ that is $65 - (7 + 21 + 20 + x)$ and using the fourth row, the unfilled box will have $x - 1$ that is $65 - (16 = 17 - x + 10 + 23)$. continuing in this fashion, we shall fill the table as shown above and considering the fourth column, we have $18 + y + x - 2 + 10 + 24 = 65$.
   Hence $x + y = 65 - 40 = 15$.

8. For the given equations
$$\begin{aligned} x+y &= 2m^2 \\ y+z &= 6m \\ x+z &= 2 \end{aligned}$$
sum of the equations gives $x = y+z = m^2+3m+1$. Therefore
$x = (x+y+z)-(y+z) = m^2-3m+1$, $y = (x+y+z)-(x+z) = m^2+3m-1$ and $z = (x+y+z)-(x+y) = -m^2+3m+1$
The values of m is $x \le y \le z$ is the same as $m^2-3m+1 \le m^2+3m-1 \le -m^2+3m+1$.
For $x \le y$, we have $m^2-3m+1 \le m^2+3m-1$ or $m \ge \frac{1}{3}$.
For $y \le z$, we have $m^2+3m-1 \le -m^2+3m+1$ or $m^2 \le 1 \Longrightarrow -1 \le m \le 1$. Hence the values of m lie in the range $\frac{1}{3} \le m \le 1$

9. In the first race, Jane ran $100m$ in the same time Sarah ran $95m$. This means the ratio of their speeds is $100:95 = 20:19$. In the time Sarah runs $1m$, Jane runs $\frac{20}{19} = 1.053m$ or in the time Jane runs $1m$, Sarah runs $\frac{19}{20} = 0.95m$. n the second race, Jane must run $105m$ and Sarah must run $100m$. If Sarah finishes first, then Jane must not have completed 105 in the time that it takes Sarah to complete $100m$. Since Jane runs $1.053m$ for $1m$ that Sarah runs, then Jane will run $105m$ in the time Sarah runs $100m$. This implies Jane must finish first. When Jane runs $105m$, Sarah will run $105 \times \frac{9}{20} = 99\frac{3}{4}m$. At this time, Sarah was $100 - 99\frac{3}{4} = \frac{1}{4}m$ behind Jane when she crossed the finish line.

10. $f : \mathbb{R}^+ \longrightarrow \mathbb{R}$ such that $xy\left(3f(\frac{x}{y}) + 5f(\frac{y}{x})\right) = 3x^2+5y^2$

Divide by xy, the equation becomes $\left(3f(\frac{x}{y}) + 5f(\frac{y}{x})\right) = 3\frac{x}{y} + 5\frac{y}{x}$

put $y = 1$ and multiply the equation by 3 gives $9f(x) + 15f(\frac{1}{x}) = 9x + 15\frac{1}{x} \cdots\cdots\cdots$ (i)
putting $x = 1$ and $y = x$ and multiply by 5 gives $15f(\frac{1}{x}) + 25f(x) = 15\frac{1}{x} + 25x \cdots\cdots$ (ii)
solving equations (i) and (ii) simultaneously gives $15f(\frac{1}{x}) + 25f(x) = 15\frac{1}{x} + 25x$
$(25-9)f(x) = 25x - 9x \Longrightarrow 16f(x) = 16x$.
Hence $f(x) = x$ for all x.

SECTION B : 10 marks each

11. As the boxes are different, we can give a label to them and deal with them in order. The first box gets 3 of the balls- this can be done in

12. Let x be the number on the second card. Then, the number on the third card is $20-2-x = 18-x$. The number on the fourth card is $20-x-(18-x) = 2$.
The number on the fifth card is $20-(18-x)-2 = x$ again. This implies the pattern repeats in the cycles of 3. The pattern is shown below $2, x, 18-x, 2, x, 18-x$. This follows that the number on the 52nd is the same as the number on the third card. so, $8 = 18-x \Longrightarrow x = 10$

13. If a, b, c form a geometric sequence then, $\frac{b}{a} = \frac{c}{b} \Longrightarrow b^2 = ac \cdots\cdots$ (i)
$\log_6 a + \log_6 b + \log_6 c = 6 \Longrightarrow \log_6(abc) = 6 \Longrightarrow abc = 6^6$
From (i), $b^3 = 6^6 = (6^2)^3$. Thus $b = 36$.

If $b-a$ is a sequence of a natural number then the possible choices are $\{11, 20, 27, 32, 35\}$.

14. let P, Q and R be real numbers

$$\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\ldots+\frac{n}{2^n}=\frac{Pn+Q}{2^n}+R$$

If it is true for $n=n+1$, then

$$\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\ldots+\frac{n}{2^n}+\frac{n+1}{2^{n+1}}=\frac{P(n+1)+Q}{2^{n+1}}+R$$

$$\frac{Pn+Q}{2^n}+\frac{n+1}{2^{n+1}}=\frac{P(n+1)+Q}{2^{n+1}} \Longrightarrow 2(Pn+Q)+(n+1)=P(n+1)+Q$$

$$2Pn+2Q+n+1=nP+P+Q \Longrightarrow n(P+1)+(Q+1-P)=0$$

on equating coefficients, we get $P+1=0 \Longrightarrow P=-1$ and $(Q+1+1)=0 \Longrightarrow Q=-2$.
Therefore $P+Q+R=-1-2+R$.
If $n=1$ in the first equation , then we have that $R=2$.
Thus $P+Q+R=-1-2+2=-1$

15. Area of triangle ABC $=\frac{1}{2}vw=\frac{1}{2}xy \Longrightarrow xy=vw$. By Pythagoras theorem, $w^2+v^2=x^2$.
Consider $35^2=(v+w)^2=v^2+w^2+2vw=v^2+w^2+2xy=x^2+2yx=x^2+2yx+y^2-y^2$.
$\Longrightarrow 35^2=(x+y)^2-y^2=(37)^2-y^2 \Longrightarrow y=12$.
Thus, $y+8=12+8=20$